Session plan 4: Data and measurement

Thu 08.10.26, 14:15–15:45 · Room 4.B51 · Basic reading: KW ch. 5 (pp. 104–124) · Applied reading: Little & Meng (2023), *Measuring democratic backsliding* · Presenters: **Nuraida Abdykapar Kyzy, Simon Ferdinand Eibach, Martins Titilola**

Learning goals: students can (1) tell a concept from a variable and explain operationalization; (2) define reliability and face, content and construct validity; (3) explain why more data reduces random error but not bias.

Before class: presentation slides in OLAT (due Wed 07.10, 12:00)? · seminar questions 3 and 4 uploaded? · grading sheet printed (`00_grading/presentation_grading_sheet.pdf`) · count of missing R tracks (`00_admin/track_choices.csv`) · deck in speaker view (press S)

Checkpoints: 14:43 start "Assessing validity" · **15:05 end group exercise** · **15:08 presentation starts** (25 min max., then discussion until 15:42)

Time	Slide	Content	Interaction	Guiding questions	If behind
14:15–14:18	2	Recap session 3: motivating question → theory → model → **hypothesis**; theories relate concepts, hypotheses relate variables	Input	Where did we stop last time?	
14:18–14:22	3	Concepts are abstract and not directly observable; example political sophistication (KW p. 110)	Question to the room	Can you *see* political sophistication? What do you see instead?	
14:22–14:27	4	Operationalization = concept → variable (KW p. 105); even income is tricky: student with wealthy parents earns zero (KW pp. 111–112)	Input + question	Is income easy to measure? What would you ask instead?	
14:27–14:29	5	KW Fig. 1.2: top row concepts, bottom row variables; KW say "variable" for both, we keep it for what we measure	Input		
14:29–14:33	6	Reliability (bathroom scale, coders); validity: face, content, construct (KW pp. 112–115)	Input	Scale shows 150, then 146: unreliable? And 157 a week later?	
14:33–14:39	7	**Target quiz**: labels hidden, revealed on click; then valid ≈ low bias, reliable ≈ low variance	Whole-class quiz	Which target is valid but not reliable? Reliable but not valid? Which is a scale that always adds 5 kg?	
14:39–14:43	8	Simulation: running mean settles on the truth with random error, on the wrong value with bias; KW: biased but reliable beats unreliable for theory tests	Input	Would 10,000 more measurements fix a biased scale?	
14:43–14:50	9	Validity types in depth: Polity has face validity (Brazil) but lacks content validity (no participation, US = 10); tolerance: Stouffer vs. Sullivan (KW pp. 116–122)	Input + question	What does Polity leave out? Why does "tolerance presupposes opposition" change the findings?	Skip tolerance, keep Polity
14:50–14:53	10	Callback session 3: Turnbull-Dugarte & Wagner measure projection with a forced choice between fictional characters	Input	How do you measure something people would not tell you?	One sentence

Time	Slide	Content	Interaction	Guiding questions	If behind
14:53–15:05	11–12	**Pair exercise** on seminar questions 3 and 4: refine RQ and hypothesis, improve indicators	Pairs (~7 min) + 1–2 pairs report (~5 min)	Is there an indicator for every concept? Does it cover the whole concept? Would someone else measure it the same way?	Cut reporting; **hard stop 15:05**
15:05–15:07	13–14	KW Fig. 5.1: reliable? then valid? Only both are useful; measurement needs theory (KW pp. 122–123)	Input		
15:07–15:08	15	Hand over to the presenters			
15:08–15:33	—	**Student presentation** (25 min max.): summary and critique ~10 min, alternative design ~15 min	Presentation; grade with the sheet		Signal at 20 min
15:33–15:42	—	**Discussion** to improve the proposal (prompts below)	Plenary	See prompts below	
15:42–15:45	16–19	Session 5 at 16:15, same room (KW ch. 6; Hünerbach & Irving on Barnes & Karim); missing R tracks (no names)	Admin		

Discussion prompts (if the room is quiet): Is an "objective" indicator unbiased? Incumbents can also lose for performance reasons (p. 158). · Are elections and turnover all of democracy (content validity)? · Can coder bias change over time, and why does that matter more than constant bias (KW p. 114 vs. Little & Meng p. 152)? · Population weights or unweighted country averages (p. 151)? · What would the presenters' alternative design measure, and how reliably?

Grading (0–5 each, 15/30 = pass): clarity · understanding · critique quality · reasoning · feasibility · discussion. Three presenters, one grade.

Notes during class: